PRACTICE TEST 2, 2025-2026

CLASS- X

SUBJECT- SCIENCE (BIOLOGY)

ANSWER KEY

Section- A

1. A student performs an experiment using a Balsam plant with intact roots, stem, leaves and flowers. The plant was kept in a test tube containing water mixed with eosin (a pink colour stain). The test tube mouth was covered using a cotton plug. After 2-3 hours, a transverse section of the stem was obtained and studied under microscope. The study revealed presence of pink colour in the xylem.

What does this observation explain? (1)

(a) Most portion of the plant stem is occupied by xylem.

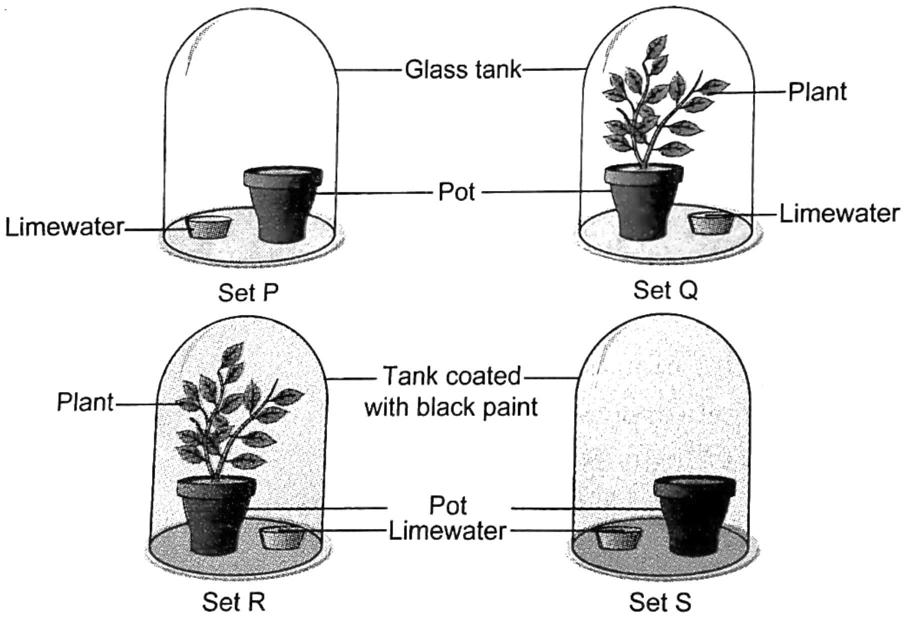
(b) Eosin solution gets stored in the phloem.

(c) Water moves through xylem in plants.

(d) None of the above.

Ans- (c) Water moves through xylem in plants.

2. Limewater turns milky in the presence of a gas which is produced during respiration. Shown below are four setups kept in sunlight for 8 hours.



In which setup is limewater expected to turn milky the most? (1)

(a) Setup P

(b) Setup Q

(c) Setup R

(d) Setup S

Ans- (c) Setup R

3. Which of the following statements are true about the brain?

(i) The main thinking part of the brain is hind brain.

(ii) Centres of hearing, smell, memory, sight, etc. are located in fore brain.

(iii) Involuntary actions like salivation, vomiting, blood pressure are controlled by the medulla in the hind brain.

(iv) Cerebellum does not control posture and balance of the body. (1)

(a) (i) and (ii)

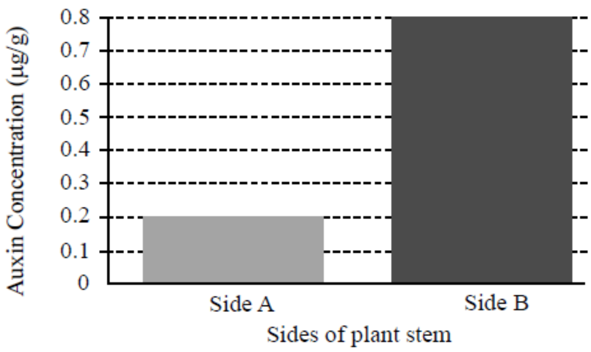
(b) (i), (ii) and (iii)

(c) (ii) and (iii)

(d) (iii) and (iv)

Ans- (c) (ii) and (iii)

4. A study records the concentration of auxin in different sides of a plant stem. The auxin concentrations are represented in the graph below.



Based on the graph, what is the likely outcome of this auxin distribution on the plant's growth? (1)

(a) The plant will grow uniformly

(b) The plant will bend towards side A

(c) The plant will bend towards side B

(d) The plant will stop growing

Ans- (b) The plant will bend towards side A

5. In cattle, having horns (h) is a recessive trait and not having horns (H) is dominant. When horned cattle are crossed with hornless cattle, the number of offspring having horns was equal to those not having horns.

Which of the following is most likely to be true? (1)

(a) Both the parents are homozygous dominant

(b) One parent is heterozygous dominant

(c) Both the parents are heterozygous

(d) Both the parents are recessive

Ans- (b) One parent is heterozygous dominant

6. A food web is considered more stable than a food chain because a food web (1)

(a) transfers all of the energy to the herbivores

(b) includes only consumers of diverse types

(c) is simpler than a food chain

(d) includes alternative pathways for energy flow

Ans- (d) includes alternative pathways for energy flow

7. In 1987, an agreement was formulated by the United Nations Environment Programme (UNEP) to freeze the production of X to prevent the depletion of Y. X and Y respectively referred here are: (1)

(a) Ozone; CFCs

(b) CFCs; UV rays

(c) CFCs; Ozone

(d) Ozone; UV rays

Ans- (c) CFCs; Ozone

The following two questions (Q.8 and Q.9) consist of two statements - Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below:

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, and R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

8. Assertion (A): In a few reptiles, the temperature at which the fertilized eggs are kept decides the sex of the offsprings.

Reason (R): Sex is not genetically determined in these animals. (1)

Ans- (a) Both A and R are true, and R is the correct explanation of A.

9. Assertion (A): We do not clean natural ponds but an aquarium needs to be cleaned regularly.

Reason (R): Natural decomposers are absent in an aquarium. (1)

Ans- (a) Both A and R are true, and R is the correct explanation of A.

10. A person suffering from liver disease is advised to avoid fatty and highly acidic foods. Comment upon the statement with justification. (2)

Ans- Liver secretes bile juice which helps in emulsification of fats and making the acidic food alkaline to facilitate further digestion. Secretion of bile juice may be hampered in a person suffering from liver disease. As a result, digestion of fat and acidic food will be slow.

11. ***Students to attempt either option A or B.***

A. Leaves of a healthy plant were coated with vaseline. Will this plant remain healthy for long? Give reason for your answer. (2)

OR

B. Why do you think that arteries need thick walls while veins do not need thick walls? (2)

Ans-

A. No, the plant will not remain healthy for long. This is because, the vaseline coated on the surface of the leaves would block the stomata and prevent loss of water through transpiration. Exchange of gases takes place through stomata which will also be hindered.

OR

B. Arteries receive the blood that emerges from the heart under high pressure, so they have thick walls. Veins do not need thick walls as they collect the blood from different organs which is no longer under high pressure.

12. (a) Construct a food chain with the following organisms:

Grasshopper, Hawk, Grass, Snake, Frog

(b) In the food chain constructed by you, if 1000 J of energy is available to frogs, how much energy will be available to hawk? (2)

Ans- (a) Grass → Grasshopper → Frog → Snake → Hawk

(b) Energy available to frog= 1000 J

Energy available to snake= 10% of 1000 J = 100 J

Energy available to hawk= 10% of 100 J = 10 J

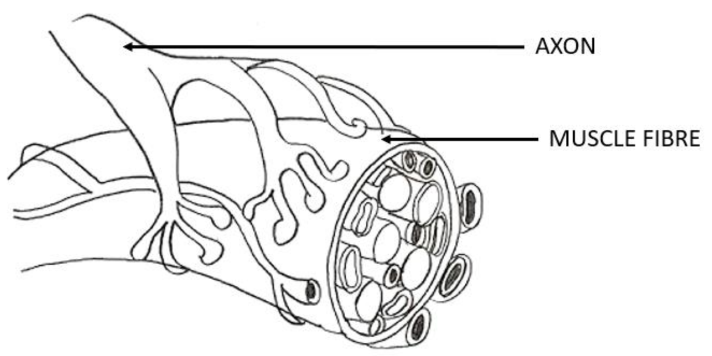
13. The nerve impulse gets delivered from the axon ending of a neuron to the muscle fibre at the neuromuscular junction.

(a) Draw a neat diagram of a neuromuscular junction and label the two components of the region.

(b) Briefly explain how are the impulses transmitted from the neuron to the muscle fibre at such a neuromuscular junction. (3)

Ans-

(a)



NEUROMUSCULAR JUNCTION

(b) At the end of the axon, the electrical impulse sets off the release of some chemicals. These chemicals cross the gap between the axon ending and muscle fibre at the neuromuscular junction and start a similar electrical impulse in the muscle fibre.

14. Sahil performed an experiment to study the pattern of inheritance of genes in pea plants. He crossed tall plants with round seeds (TTRR) and dwarf plants with wrinkled seeds (ttrr) and obtained all tall plants with round seeds in the F_1 generation.

(a) What will be the genotype of the plants in the F_1 generation? Support your answer with cross.

(b) If 1600 plants were obtained in the F_2 progeny after self-pollinating the plants of the F_1 generation, write the number of plants which are tall and with wrinkled seeds. Justify your answer with cross. (3)

Ans- (a) The genotype of the plants in F_1 generation is TtRr.

P gen… Tall plants with round seeds ⊗ Dwarf plants with wrinkled seeds

TTRR ⊗ ttrr

↓ ↓

G_1 TR tr

F_1 TtRr

b) F_2 generation is shown with help of Punnett square below:

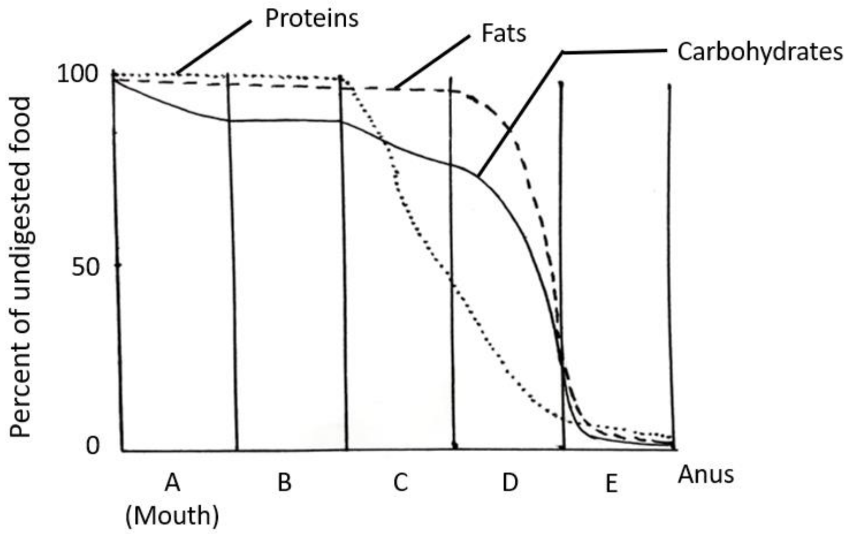
♀ \ ♂	TR	Tr	tR	tr
TR	TTRR Tall round	TTRr Tall round	TtRR Tall round	TtRr Tall round
Tr	TTRr Tall round	TTrr Tall wrinkled	TtRr Tall round	Ttrr Tall wrinkled
tR	TtRR Tall round	TtRr Tall round	ttRR Dwarf round	ttRr Dwarf round
tr	TtRr Tall round	Ttrr Tall wrinkled	ttRr Dwarf round	ttrr Dwarf wrinkled

Out of 16 combinations, 3 plants are tall with wrinkled seeds.

So, number of plants which are tall and with wrinkled seeds= (3/16) x 1600 = 300.

15. **Read the passage carefully and answer the following questions:**

In humans, the process of nutrition takes place through a long tube extending from the mouth to the anus along with various accessory organs and glands. The graph below shows the extent to which carbohydrates, proteins and fats are chemically digested as the food passes through the human digestive tract. The vertical axis indicates the percentage of undigested food remaining as the food moves down through the digestive tract. The letters on the horizontal axis represent the various parts of the digestive tract.



Attempt either subpart A or B

A. Based on the above graph, how can you tell that there is no digestion taking place in part B? Name the part of the digestive tract represented by B.

OR

B. Based on the above graph, which part of the digestive tract would have the greatest amount of the digested food? Give reason for your answer.

C. Why is there a sharp decrease in the percent of undigested proteins in part C?

D. If a person eats roti for dinner, based on the above graph, at which part of the digestive tract will it start getting digested? Name the enzyme that facilitates this digestion. (4)

Ans-

A. In part B, there is no decline in the percent of undigested food for all the three types of nutrients. Hence, we can tell that there is no digestion taking place in part B.

Part B represents oesophagus.

OR

B. Part D (Small intestine).

This is because part D in the graph shows maximum decline in the percent of undigested food for all the three types of nutrients.

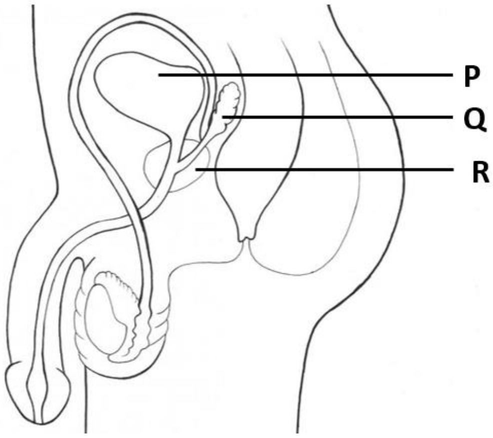
C. Part C is stomach in which a protein digesting enzyme pepsin is secreted.

D. Part A.

Salivary amylase / Ptyalin is the enzyme that facilitates digestion.

16. ***Attempt either option A or B.***

A. Humans use sexual mode of reproduction. The transfer of germ cells between two people needs special organs for the sexual act. The figure given below shows the human male reproductive system.



a) In human males, why are the testes located outside the abdominal cavity?

b) The testes secrete a hormone X which regulates the formation of sperms. Identify X. State another function of X apart from the one mentioned in the question.

c) Parts Q and R marked in the figure add their secretions to the sperms move through the vas deferens. What advantages do these secretions provide to the sperms? (5)

Ans-

a) The testes are located outside the abdominal cavity to maintain the temperature required for production of sperms which is about 2-3°C lower than that of the body temperature.

b) X- Testosterone.

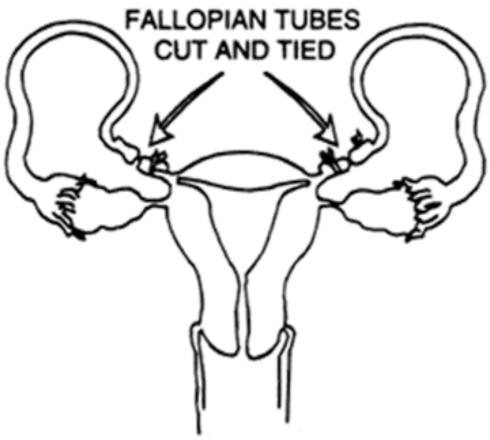
Testosterone stimulates the formation of secondary sexual characters in males at puberty.

c)

- Provides a medium for the transport of sperms
- Provides nutrition to the sperms

OR

B. The sexual act is a very intimate connection of bodies, so it may lead to transmission of many diseases. Moreover, the sexual act also has the potential to lead to pregnancy. As reproduction leads to increase in population size, the population explosion has become a growing concern for the society. Various methods of family planning should be adopted. One such contraceptive method is shown in the figure below.



a) A surgical contraceptive method is shown in the figure above. How do you think this procedure helps to avoid pregnancy?

b) Certain drugs which need to be taken orally as pills can act as contraceptive. How do these drugs function?

c) Name two sexually transmitted diseases caused by bacteria. Mention one way to prevent the transmission of such diseases during the sexual act. (5)

Ans- a) Prevents fertilisation to take place.

b) The drugs change the hormonal balance of the body so that eggs are not released and fertilisation cannot occur.

c) Gonorrhoea and syphilis.

By using a covering, called a condom, on the penis during sexual intercourse helps to prevent transmission of such diseases.
